Enhancing Foraging Safety: A Low-Resource Machine Learning Model for Poisonous Mushroom Identification in the Philippines

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ABSTRACT

The identification of toxic species of wild mushrooms in the Philippines remains a persistent public health risk, often resulting in avoidable poisoning cases. The study developed a mobile prototype based on low-resource machine learning, capable of differentiating between three toxic species or presumed toxic species, including *Chlorophyllum molybdites*, *Lepiota cristata*, and *Agaricus xanthodermus*, from their edible look-alikes *Chlorophyllum rhacodes, Macrolepiota procera*, and *Agaricus campestris*. Using a curated dataset of 58,184 images across six classes sourced from iNaturalist, transfer learning was applied to fine-tune lightweight convolutional neural network architectures and a Vision Transformer architecture for a hybrid offline-online application. Model evaluation prioritized recall, F-1 score, and confusion matrix analysis over standard accuracy to minimize false negatives, wherein toxic species could be misclassified as edible. MobileNetV4-Large was selected as the offline model, which achieved an average latency of 5.05 seconds on the selected low-end Android device. Meanwhile, Swin Transformer Base was chosen for online use, and it attained a macro F1-score of 93.24% alongside the lowest false-negative rate among toxic classes. Independent machine learning experts reviewed the methodology and model development process. End-user testing with 20 participants was conducted through purposive sampling and a modified PSSUQ-based instrument, which yielded an average score of 4.72 out of 5. These findings suggested that low-resource machine learning models can improve mushroom foraging safety in resource-constrained settings.

CCS CONCEPTS • Computing methodologies → Computer vision; Neural networks; Transfer learning • Human-centered computing → Mobile computing • Applied computing → Health informatics

Keywords: Mushroom poisoning identification, transfer learning, Vision Transformer, lightweight CNN, mobile deep learning application, MobileNetV4, Swin Transformer

1. INTRODUCTION

The tropical archipelago of the Philippines is a global biodiversity hotspot where wild mushrooms thrive abundantly during the rainy season, with recent reviews cataloging at least 447 distinct species from 2001 to 2021 (Dulay et al., 2023). While physical characteristics such as cap shape, gill color, bruising, and stalk structure can be used to identify species and toxicity (Ortiz-Letechipia et al., 2024), misidentification of wild mushrooms contributes to recurring poisoning incidents. News reports, including accounts sourced from the Department of Health (DOH), document 34 hospitalizations across Benguet and Mountain Province between May and June 2024 (Philippines News Agency, 2024) and an earlier outbreak involving up to 14 individuals (Rendon, 2017). More recently, 10 residents, including seven children, were hospitalized in Talibon, Bohol, in March 2026 after eating unspecified poisonous mushrooms (Inojales, 2026).

In tropical regions such as the Philippines, *Chlorophyllum molybdites* (False Parasol or Green-Spored Parasol) is a commonly reported culprit, closely resembling generally edible parasol mushrooms such as *Macrolepiota procera* (Parasol Mushroom) and *Chlorophyllum rhacodes* (Shaggy Parasol). Similarly, *Lepiota cristata* (Stinking Dapperling) shares morphological traits with young edible parasols. Its toxicity is not firmly established, as an analysis of Turkish specimens found no amatoxins or phallotoxins, though the authors cautioned this does not make the species edible (Yılmaz et al., 2018). Because other small Lepiota species are dangerous, including *L. brunneoincarnata*, which has caused severe liver injury requiring transplantation (Anbardar et al., 2024), and *L. spiculata*, which contains lethal α-amanitin (Angelini et al., 2020), L. cristata was treated as a precautionary toxic class. Another widespread hazard is *Agaricus xanthodermus* (Yellow Stainer), reported by iNaturalist contributors to account for roughly half of poisoning cases among Agaricus-type foragers (iNaturalist, n.d.), and frequently mistaken for the edible *Agaricus campestris* (Field Mushroom).

Traditional identification relies on field guides, spore prints, or mycological consultation, which are largely inaccessible to non-specialists, indigenous foragers, and rural communities. While computer vision and deep learning have advanced automated species identification globally, existing applications appear to give limited attention to tropical look-alike hazards or to low-resource smartphones in low-connectivity settings. This study addresses this gap by developing and evaluating a hybrid offline-online mobile prototype built on transfer learning. The system integrates MobileNetV4-Large via ExecuTorch for fully offline on-device inference and Swin Transformer Base for online classification, together with a three-view probability-averaging mechanism and a heuristic three-tier confidence warning system intended to support safer foraging decisions. The remainder of this paper presents the related literature, methods, results, and discussion of the prototype's evaluation.

2. REVIEW OF RELATED LITERATURE

2.1 Philippine Mycological Biodiversity and Toxic Look-Alikes

Philippine mycological checklists compiled by Dulay et al. (2023) and Jacob et al. (2022) confirm the local occurrence of toxic species such as *Chlorophyllum molybdites*, *Agaricus xanthodermus*, and *Lepiota cristata*, and document severe gastrointestinal symptoms associated with mushroom species found in the country. However, these checklists note a lack of dedicated profiling and public-facing tools that help non-specialists distinguish toxic species from their edible look-alikes.

2.2 Deep Learning in Macrofungi Classification

Ozsari et al. (2024) evaluated deep learning architectures for classifying macrofungi species, including *Amanita pantherina*, *Boletus edulis*, and *Pleurotus ostreatus*, and reported performance differences among the architectures. In Southeast Asia, Lim et al. (2024) curated a local Malaysian dataset and compared Vision Transformers (ViTs) and ResNet variants for mushroom identification, addressing poisoning risks from outdoor activities in humid climates similar to those in the Philippines. Locally, Algarme and Villegas (2025) explored machine learning for conservation and health awareness in the Philippines, but their work focused on broader mushroom profiling rather than toxic look-alike differentiation or mobile deployment.

2.3 Lightweight CNNs for Edge Deployment

Deploying deep learning on resource-constrained mobile and edge devices requires computationally and memory-efficient models (Cai et al., 2022). MobileNetV3-Large was developed using hardware-aware Neural Architecture Search and the NetAdapt algorithm to optimize performance on mobile hardware, achieving a 3.2-percentage-point ImageNet top-1 accuracy improvement over MobileNetV2 while reducing latency by up to 20% (Howard et al., 2019). Zhu et al. (2023) applied an improved MobileNetV3-Large-based model to classify shiitake mushroom quality against complex backgrounds, achieving 99.91% accuracy with a relatively small model size. Their task differs from toxic-versus-edible classification, but it supports the suitability of the architecture for mushroom imagery. MobileNetV4 is a later generation of the family. Qin et al. (2024) introduced the Universal Inverted Bottleneck and Mobile MQA attention, reporting a 39% speedup under their evaluated conditions across mobile CPUs, GPUs, DSPs, and accelerators such as the Apple Neural Engine and Google Pixel EdgeTPU. This efficiency across hardware makes it relevant for offline deployment on low-end Android devices.

2.4 High-Capacity CNN and Vision Transformer Architectures

Where network connectivity permits server-side inference, higher-capacity models can be used. ConvNeXt-Base modernizes standard convolutional networks using design principles from Vision Transformers while remaining purely convolutional, reaching 83.8% top-1 accuracy on ImageNet-1K with high inference throughput (Liu et al., 2022). Swin Transformer Base applies hierarchical self-attention within shifted local windows, allowing information to be exchanged across image regions at manageable computational cost, and reaches 83.5% top-1 accuracy on ImageNet-1K (Liu et al., 2021). Its attention-based design offers a contrasting approach to ConvNeXt-Base for distinguishing visually similar classes.

2.5 Confidence Estimation and System Usability Validation

Corbière et al. (2021) argued that reliable confidence estimates are required for deploying classifiers in safety-critical applications and described withholding predictions when confidence is low (a reject option) as a way to reduce the risk of unreliable outputs. For usability validation, Vlachogianni and Tselios (2023) found that mobile applications evaluated with the Post-Study System Usability Questionnaire (PSSUQ) scored higher on user satisfaction and system usability than other technology categories, supporting its use for evaluating mobile applications.

3. CONCEPTUAL FRAMEWORK

This study was guided by an Input-Process-Output (IPO) conceptual framework illustrating how raw citizen-science mushroom observations were transformed into an evaluated hybrid mobile identification tool (Figure 1).

The Input phase comprises the curated image dataset of six wild mushroom species (three toxic or presumed-toxic species and their three edible look-alikes) sourced from the iNaturalist API (Research Grade observations only), pre-trained CNN and Vision Transformer checkpoints, and the evaluation instruments (a modified PSSUQ questionnaire and expert validation sets).

The Process phase covers data cleaning and group-aware splitting to prevent specimen-level leakage; image preprocessing (resizing to 384×384 pixels, normalization) and augmentation via Albumentations; full fine-tuning through transfer learning across four candidate architectures; model evaluation prioritizing recall, macro F1-score, and false-negative minimization; integration into a Flutter/ExecuTorch hybrid mobile application with three-view probability averaging and heuristic confidence thresholds; and expert validation with Central Luzon State University (CLSU) mycologists and machine learning specialists, followed by end-user testing.

The Output phase delivers the evaluated hybrid Android prototype, comparative model performance metrics, expert validation results, and PSSUQ usability ratings.

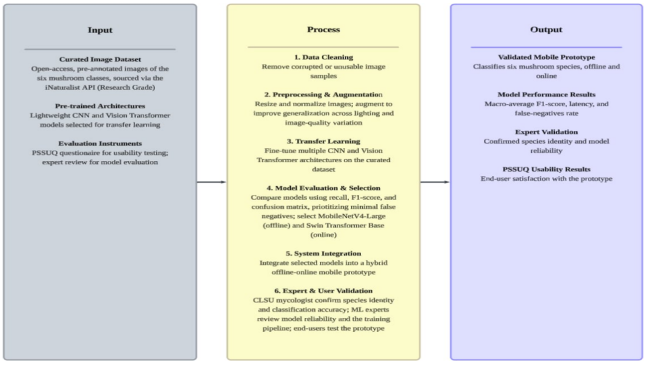


Figure 1: Input-Process-Output conceptual framework for the low-resource mushroom identification system.

4. OBJECTIVES OF THE STUDY

This study aimed to develop and evaluate a machine learning prototype for identifying select poisonous wild mushrooms and their edible look-alikes documented to occur in the Philippines, as an accessible hybrid offline-online tool for both low-connectivity and connected settings. Specifically, the study aimed:

1. To curate and transform an image dataset of six wild mushroom species (toxic *Chlorophyllum molybdites* and *Agaricus xanthodermus*, presumed-toxic *Lepiota cristata*, and edible *Chlorophyllum rhacodes*, *Macrolepiota procera*, and *Agaricus campestris*) using group-aware splitting and data augmentation.
2. To train and fine-tune lightweight CNN architectures (MobileNetV3-Large, MobileNetV4-Large) for offline on-device use and higher-capacity architectures (ConvNeXt-Base, Swin Transformer Base) for online inference.
3. To evaluate and compare the models using recall, macro F1-score, and confusion matrix error analysis, with accuracy and on-device latency on a low-end Android smartphone as supporting measures.
4. To assess the system through independent mycological identification, machine learning expert review, and end-user usability testing with a modified PSSUQ-based instrument, using purposive sampling of mushroom enthusiasts, environmental groups, ML/AI experts, and the general population.

5. METHODS

5.1 Research Design

This study used a quantitative design complemented by qualitative expert consultations. System development followed a prototyping approach across three iterations, each incorporating requirements, design, build, user evaluation, and refinement. The machine learning pipeline followed a Low-Resource Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology (Figure 2).

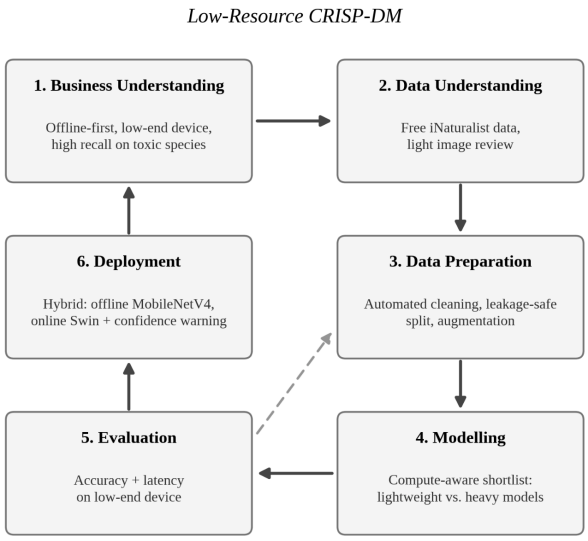


Figure 1: Low-Resource CRISP-DM methodology model.

5.2 Dataset Curation and Preprocessing

Images were collected through a custom Python pipeline using the iNaturalist API (v1/observations), restricted to Research Grade observations released under CC-BY, CC-BY-NC, or CC0 licenses. A total of 59,315 raw images were retrieved across the six classes. Manual quality assurance removed 1,131 images that were inconclusive, microscopic, dried or dead, or contained text, leaving 58,184 images.

To eliminate specimen-level leakage, the dataset was re-split with a group-aware method so that all photographs of one specimen fell in the same partition. The resulting 80/10/10 split was 46,474 training (79.87%), 5,897 validation (10.14%), and 5,813 test (9.99%) images. Images were resized to 384×384 pixels and normalized. Training used Albumentations (RandomResizedCrop, flips, random 90-degree rotations, Gaussian blur, color jitter, coarse dropout) together with batch-level Mixup and CutMix.

5.3 Model Training and Platform Configuration

Models were trained in Python 3.13.15 with PyTorch 2.11.0+cu128 on a Google Colab runtime with an NVIDIA L4 GPU, an Intel Xeon CPU at 2.20 GHz, and 52.96 GB RAM. Four pre-trained timm checkpoints were fully fine-tuned under one 15-epoch pipeline: mobilenetv3_large_100.miil_in21k_ft_in1k, mobilenetv4_conv_aa_large.e230_r384_in12k_ft_in1k, convnext_base.fb_in22k_ft_in1k_384, and swin_base_patch4_window12_384.ms_in22k_ft_in1k. Training used AdamW with an initial learning rate of 5×10^{-5}, weight decay of 0.05, layer-wise learning rate decay (0.85), cosine annealing with warmup, stochastic depth (0.2), class-weighted cross-entropy with label smoothing (0.1), and an Exponential Moving Average (EMA) of weights.

5.4 System Architecture and Inference Pipeline

The hybrid Android prototype was built with Flutter and Dart. Offline mode ran MobileNetV4-Large locally through ExecuTorch in FP32, while online mode sent requests to Swin Transformer Base

hosted on Google Cloud. Because a single view is often non-diagnostic, users submit three photographs per specimen: top (cap), bottom (gills), and side (stalk). Each image yields a six-class softmax vector, and the system averages the three: $P_fused(c) = (1/3)\ \Sigma_i\ P_i(c)$, for views $i \in \{top, bottom, side\}$. The class with the highest fused probability is selected.

A heuristic three-tier confidence system manages uncertainty. Predictions with ≥85% confidence are shown directly (red for toxic, green for edible). Scores of 50% to 84% trigger a yellow cautionary warning advising expert verification. Scores below 50% are withheld as "Image Unclear / Inconclusive."

5.5 Validation and Evaluation Metrics

Models were evaluated on the held-out test set (N = 5,813) with emphasis on toxic-class recall, since a toxic specimen misread as an edible look-alike is the costliest error. The study set a target of at least 85% recall and macro F1-score. Metrics were computed in Python with Scikit-Learn and NumPy.

5.5.1 **Accuracy**

The proportion of correct classifications across all six classes, Accuracy = (TP + TN) / (TP + TN + FP + FN), reported on the test set (N = 5,813) and the CLSU expert-confirmed set (N = 35).

5.5.2 **Precision**

The share of predictions for a class that truly belong to it, Precision = TP / (TP + FP), macro-averaged so all six classes count equally.

5.5.3 **Recall**

The share of actual specimens of a class that the model found, *Recall = TP / (TP + FN). For the toxic classes this directly measures the rate of dangerous false negatives.*

5.5.4 **F1-Score**

The harmonic mean of precision and recall, F1 = 2 × (Precision × Recall) / (Precision + Recall), macro-averaged across classes and used as the primary metric for comparing architectures. Confusion matrices were used to examine the direction of errors between toxic species and their edible look-alikes.

On-device latency was measured over six trials per mode on a low-end Android smartphone (Qualcomm Snapdragon 450, octa-core 1.8 GHz Cortex-A53, Adreno 506 GPU, 2 GB RAM). Mycologists from Central Luzon State University (CLSU) confirmed 35 temporally separated iNaturalist specimens (105 sub-images uploaded in August, distinct from the June/July training cutoff), plus one out-of-scope specimen (*Marasmius* sp.). Two independent ML experts reviewed the training pipeline and evaluation methodology. Twenty purposively sampled participants (10 general users, 3 mushroom enthusiasts, 3 environmental group members, and 4 future researchers and technology practitioners) rated usability with a modified PSSUQ-based instrument, analyzed descriptively across the System Usefulness, Information Quality, and Interface Quality subscales.

5.6 Ethical Considerations

The study followed the Data Privacy Act of 2012 (Republic Act No. 10173) and obtained informed consent from all survey participants. All iNaturalist images were used under their Creative Commons licenses. Because mushroom misidentification carries serious health risks, the interface displays warning banners stating that the tool is a decision-support aid and not a substitute for professional mycological consultation.

5.7 Research Procedure

Model development proceeded in seven phases alongside three prototype iterations. (1) A MobileNetV3-Large baseline was trained, and a cloud-storage input bottleneck was resolved by moving data to local instance storage. (2) A unified 15-epoch full-fine-tuning pipeline (Section 5.3) was established and first benchmarked on ConvNeXt-Base. (3) A leakage audit found a 0.28% cross-split specimen overlap (161 images), so the full 58,184-image dataset was re-split by observation ID, giving zero specimen overlap. (4) MobileNetV3-Large and ConvNeXt-Base were retrained on the corrected split under the identical pipeline. (5) Swin Transformer-Base (about 87.9M parameters) was trained to compare windowed self-attention with the size-matched ConvNeXt-Base (about 88.6M). (6) MobileNetV4-Large (about 32.6M parameters) was trained to test whether a newer MobileNet generation improved on V3. (7) INT8 and FP16 post-training quantization were evaluated, and FP32 was retained for deployment. MobileNetV4-Large was selected for offline use and Swin Transformer-Base for online use.

The Flutter prototype went through three iterations. Iteration 1 offered single-image offline classification. Iteration 2 added three-view probability averaging, cloud inference, the three-tier confidence system, and color-coded results. Iteration 3, following panel recommendations, added morphological explanations of each result, clearer invalid-image messages, and changed the cautionary warning color to yellow.

5.8 Testing

Testing had four parts. First, all four models were evaluated by single-pass inference (no test-time augmentation) on the held-out test set, and three-view fusion was verified with controlled cap, gill, and stalk photographs of *C. molybdites* in both modes. Second, six functional test cases (TC_01 to TC_06) covered offline and online three-image classification, non-mushroom uploads, incomplete input (two of three views), blurry or low-light images, and out-of-scope species (*Amanita muscaria*, *Leucocoprinus birnbaumii*, *Pleurotus ostreatus*, *Schizophyllum commune*) to observe closed-set behavior. Third, six end-to-end latency trials were timed per mode on the Snapdragon 450 device. Fourth, both deployed models were tested against the 35 CLSU-confirmed specimens plus the out-of-scope *Marasmius sp.*, while 20 end users tested the app on unseen sample sets and completed the modified PSSUQ survey.

6. RESULTS

6.1 Dataset Curation and Splitting Results

Manual cleaning removed 1,131 invalid images from the 59,315 images retrieved, leaving 58,184 curated images across the six classes. Group-aware splitting then partitioned them to eliminate cross-split specimen overlap (Table 1).

Table 1: Dataset Cleaning and Group-Aware Split Distribution

Species Class	Status	Raw Images	Removed	Final
Agaricus campestris	Edible	8,213	187	8,026
Agaricus xanthodermus	Toxic	9,791	284	9,507

Chlorophyllum molybdites	Toxic	10,067	169	9,898
Chlorophyllum rhacodes	Edible	10,338	134	10,204
Lepiota cristata	Toxic*	10,971	281	10,690
Macrolepiota procera	Edible	9,935	76	9,859
Total	n/a	59,315	1,131	58,184

*Note. *Presumed toxic on a precautionary basis. The group-aware split yielded 46,474 training (79.87%), 5,897 validation (10.14%), and 5,813 test (9.99%) images.*

6.2 Model Classification Performance and Error Direction

Accuracy, macro-average recall, and macro F1-score rose steadily from the lightweight baseline to the Vision Transformer (Table 2).

Table 2: Classification Performance and Critical Misclassification Counts on the Test Set (N = 5,813)

Metric	MobileNetV3-Large	MobileNetV4-Large*	ConvNeXt-Base	Swin Transformer Base**
Accuracy	87.41%	90.80%	93.15%	93.38%
Macro Precision	87.16%	90.55%	93.02%	93.27%
Macro Recall	87.25%	90.47%	92.95%	93.23%
Macro F1-Score	87.17%	90.56%	92.98%	93.24%
FN: A. xanthodermus -> A. campestris	102	68	54	44
FP: A. campestris -> A. xanthodermus	70	80	73	68
FN: C. molybdites -> C. rhacodes	75	39	30	30
FP: C. rhacodes -> C. molybdites	53	52	50	50

*Note. *Selected offline model. **Selected online model. FN = false negative (toxic specimen classified as its edible look-alike). FP = false positive (edible specimen classified as its toxic look-alike).*

MobileNetV3-Large reached a macro F1-score of 87.17%, but its recall on the toxic *A. xanthodermus* (83.33%) and *C. molybdites* (83.37%) fell below the study's 85% target, with 102 and 75 false negatives. MobileNetV4-Large raised macro F1 to 90.56% and cleared the 85% recall target on both toxic classes (88.47% and 89.60%), cutting false negatives to 68 and 39 (Figure 3).

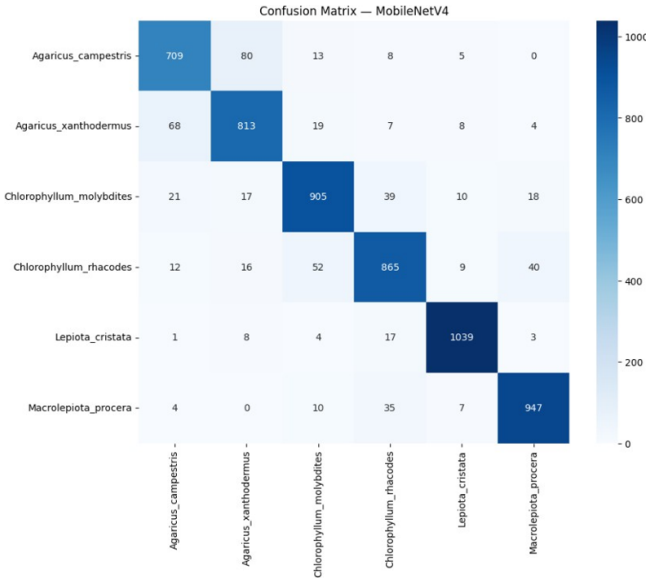


Figure 3: Confusion matrix for the selected offline model, MobileNetV4-Large.

Among the online candidates, ConvNeXt-Base reached 92.98% macro F1, and Swin Transformer Base performed best overall at 93.24% macro F1 and 93.38% accuracy. Swin's recall was 93.46% on *A. xanthodermus* and 93.27% on *C. molybdites*, with the lowest or tied-lowest toxic-to-edible false-negative counts of all models (44 and 30; Figure 4).

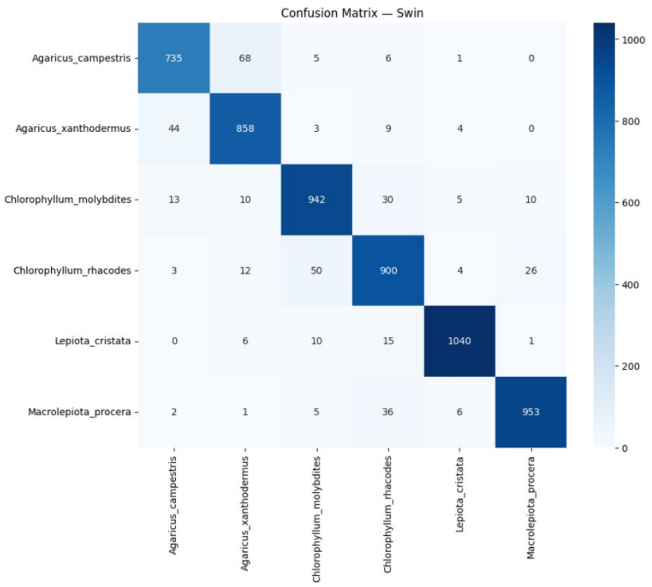


Figure 4: Confusion matrix for the selected online model, Swin Transformer Base.

6.3 System Testing, Multi-View Fusion, and On-Device Latency

All six functional test cases (TC_01 to TC_06) passed their expected outputs. Offline and online three-image classification worked, non-mushroom uploads returned a low-confidence result with an invalid-image message, and incomplete input (two of three views) disabled the submit button.

In the out-of-scope test (TC_06), four untrained species were uploaded in both modes: *Amanita muscaria* (Fly Agaric), *Leucocoprinus birnbaumii*, *Pleurotus ostreatus*, and *Schizophyllum commune* (Splitgill). Of the eight runs, three were withheld as low confidence (Fly Agaric in both modes, and Splitgill online). The other five runs were assigned to one of the six trained classes. Because this was a small, informal test (n = 4), it shows the closed-set behavior of the classifier and should not be read as a validated property of the model.

Controlled testing of three-view averaging on *C. molybdites* showed high agreement across top, bottom, and side views, with a fused confidence of 90.25% (spread 0.7%) on MobileNetV4-Large and 90.80% (spread 2.1%) on Swin Transformer Base.

On the low-end Snapdragon 450 device (2 GB RAM), six end-to-end three-view trials were run per mode (Table 3). Offline mode (MobileNetV4-Large, FP32) averaged 5,050.33 ms, and online mode (Swin Transformer Base) averaged 4,084.50 ms. The online mode was 19.1% faster on average but more variable, consistent with network conditions. Experimental FP16 quantization halved the MobileNetV4-Large file size (about 120 MB to 60 MB) with accuracy preserved within a fraction of a percentage point. However, on-device ExecuTorch inference slowed to about 78.9 seconds because of incomplete native FP16 operator support, so FP32 was retained.

Table 3: End-to-End Three-View Latency on the Selected Low-End Android Device

Mode / Model	Mean (ms)	SD (ms)	Min (ms)	Max (ms)
Offline (MobileNetV4-Large)	5,050.33	522.30	4,414	5,724
Online (Swin Transformer Base)	4,084.50	764.44	3,030	5,033

Note. Six three-image trials on a Qualcomm Snapdragon 450 (2 GB RAM) device.

6.4 Mycological Expert Validation, ML Expert Feedback, and System Usability

Against the 35 independent specimens confirmed by CLSU mycologists, MobileNetV4-Large classified 30 correctly (85.71%) and Swin Transformer Base classified 31 correctly (88.57%) (Table 4). One additional out-of-scope *Marasmius sp*. specimen returned below 50% confidence in both modes and was withheld.

Table 4: External Validation Against CLSU Mycologist-Confirmed Specimens (N = 35)

Species Class	Status	Expert Confirmed	Offline Correct	Online Correct
C. molybdites	Toxic	8	5	6
C. rhacodes	Edible	4	3	3
A. xanthodermus	Toxic	5	4	5
A. campestris	Edible	7	7	6
L. cristata	Toxic*	5	5	5
M. procera	Edible	6	6	6
Total (accuracy)	n/a	35	30 (85.71%)	31 (88.57%)

Note. Presumed toxic on a precautionary basis.

Two independent ML experts reviewed the training pipeline and evaluation. Their main points (split integrity, evaluation consistency, the definition of "low-resource," out-of-distribution handling, and prioritizing recall for the toxic classes) are addressed in the Discussion.

The 20 end users rated usability on the modified 5-point PSSUQ-based instrument (16 rated items). The overall mean was 4.72 out of 5, with Information Quality highest (4.78), then Interface Quality (4.75), and System Usefulness lowest (4.63) (Table 5).

Table 5: Modified PSSUQ Results by Participant Group and Subscale (N = 20)

Subscale	GU	ME	EG	TP	Mean
Interface Quality	4.80	4.78	4.78	4.58	4.75
Information Quality	4.80	4.73	4.87	4.70	4.78
System Usefulness	4.71	4.42	4.88	4.38	4.63
Total	4.76	4.58	4.85	4.52	4.72

Note. GU = General Users (n = 10), ME = Mushroom Enthusiasts (n = 3), EG = Environmental Groups (n = 3), TP = Future Researchers and Technology Practitioners (n = 4). Scores are on a 5-point scale.

7. DISCUSSION

7.1 Safety-Critical Metric Prioritization and Architecture Trade-offs

A central finding, reinforced during the ML expert consultations, is that overall accuracy is an insufficient primary metric for safety-critical foraging applications. Misclassifying a toxic mushroom as an edible look-alike (false negative) poses a far greater health risk than the reverse (false positive), so model selection prioritized toxic-class recall and the direction of confusion-matrix errors. Upgrading the offline model from MobileNetV3-Large to MobileNetV4-Large reduced *A. xanthodermus* false negatives by 33.3% (102 to 68) and *C. molybdites* false negatives by 48.0% (75 to 39), while keeping inference at about 5 seconds on entry-level hardware. This is consistent with the efficiency-oriented design principles described by Cai et al. (2022) and Qin et al. (2024). In online mode, Swin Transformer Base had the lowest toxic false-negative counts (44 and 30), which is consistent with the ability of shifted-window self-attention (Liu et al., 2021) to separate fine-grained morphological differences.

7.2 Edge Quantization Bottlenecks and Multi-View Fusion

FP16 quantization is expected to halve memory use and speed up inference (Taklai et al., 2025). In this study, however, running the FP16 MobileNetV4-Large model through ExecuTorch on Android took about 80 seconds per identification, which we attribute to upcasting and downcasting overhead for operators without native FP16 support. This may reflect the specific runtime and hardware used rather than a shortcoming of FP16 itself. Retaining FP32 gave a practical latency of about 5.05 seconds for three images. Requiring top, bottom, and side views also addresses the weakness of single-cap photographs, since ambiguous features in one view can be balanced by diagnostic features (such as gill attachment or a stalk ring) in another. Because the models were trained on single images, the fusion step is post-hoc averaging, and its benefit was verified only in a controlled test.

7.3 External Generalization and User Trust

On the 35 CLSU-confirmed specimens, accuracy was about 5 percentage points below test-set accuracy (85.71% offline vs. 90.80%; 88.57% online vs. 93.38%). Possible reasons include the temporal separation from the training data and the use of three-view fused inputs, but with only 35 specimens this gap should be read cautiously.

In the PSSUQ evaluation, overall satisfaction was high (4.72 out of 5). Mushroom Enthusiasts (4.42) and Future Researchers and Technology Practitioners (4.38) rated System Usefulness lower than General Users (4.71) and Environmental Groups (4.88). Because the enthusiast and environmental groups had only three participants each, these differences are descriptive. The item with the most varied responses concerned comfort in relying on the app for toxicity checks (mean 4.20, range 2 to 5). This is consistent with literature linking distrust of "black-box" AI to health-critical settings (Li et al., 2021), and it motivated the morphological explanations added in the third prototype iteration.

8. LIMITATIONS

Several factors limit these findings. First, although all six species are documented in the Philippines, training images came from iNaturalist without a geographic restriction, so regional morphological variation and tropical field lighting were not isolated. Second, the model is a closed-set classifier restricted to six species. In informal testing on four out-of-scope species (*Amanita muscaria*, *Leucocoprinus birnbaumii*, *Pleurotus ostreatus*, and *Schizophyllum commune*), three of eight runs were withheld as unclear, while the other five were assigned to a trained class. Third, the 85% and 50% confidence cutoffs rely on raw, uncalibrated softmax outputs and were set by research judgment, not by empirical calibration. Fourth, the toxic status of *L. cristata* is presumed rather than toxicologically confirmed. Fifth, models were trained on single images without view-angle labels, and the three-view fusion has not been validated beyond a controlled test. Finally, results come from a single fixed data split, a small expert set (35 specimens), a purposive sample of 20 users on a modified PSSUQ instrument, and one low-end Android device. The study did not measure any effect on actual poisoning incidents.

9. CONCLUSION

This study demonstrated the feasibility of a low-resource, hybrid offline-online mobile application for distinguishing toxic wild mushrooms (*C. molybdites*, *A. xanthodermus*, and presumed-toxic *L. cristata*) from their edible look-alikes (*C. rhacodes*, *A. campestris*, and *M. procera*). From 58,184 curated images split by specimen, a unified transfer-learning pipeline produced MobileNetV4-Large for offline inference (90.56% macro F1; 5.05 s on a 2 GB RAM device) and Swin Transformer Base for online inference (93.24% macro F1). Validation with CLSU mycologists (85.71% offline, 88.57% online accuracy), two ML experts, and 20 end users (4.72 out of 5 on the modified PSSUQ) suggests that lightweight deep learning with multi-view fusion and confidence-based warnings can serve as a practical decision-support aid for foraging safety in low-connectivity settings. It is an aid alongside expert mycological consultation, not a replacement for it.

10. RECOMMENDATIONS FOR FUTURE WORK

Future work should expand the dataset to more toxic and edible Philippine macrofungi and prioritize field-verified specimens photographed in Philippine ecosystems. To address the closed-set limitation, researchers should add out-of-distribution detection or open-set recognition and apply post-hoc confidence calibration (for example, temperature scaling) to set empirically validated rejection thresholds. Models should also be trained directly on grouped multi-view inputs rather than relying on post-hoc probability averaging. Visual explanation methods such as Grad-CAM could highlight diagnostic regions on the cap, gills, and stalk and improve user trust. Finally, future development should evaluate mobile runtimes and hardware accelerators with native INT8 and FP16 support, to compress the model without a latency penalty.

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